A Novel Hybrid Ensemble Approach for Credit Card Fraud Detection on Imbalanced Data

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ABSTRACT

Credit card fraud detection has become a challenging problem for financial systems since the datasets are extremely imbalanced and the fraud patterns continue to evolve. Machine learning is commonly used to detect fraud, where customary models show a very high false-negative rate, which leads to financial losses for the financial institution. We propose an ensemble learning based fraud detection method supplemented with the recent data balancing techniques to overcome the data imbalance problem. We utilize the Random Forest, Balanced Random Forest, XGBoost and LightGBM models to perform our experiments along with the use of SMOTE and the SMOTE-Tomek sampling techniques. Experiments on a publicly available ULB credit card fraud dataset were carried out. To obtain realistic results, the original test set was used. To evaluate the results, precision, recall, F1-score, ROC Area Under the Curve (ROC-AUC) and false negative analysis (FNA) metrics were used. Experimental results prove that the proposed XGBoost model with SMOTE-Tomek and cost-sensitive learning to improve fraud recall and decrease false negatives attained better performance than baseline models while maintaining comparable precision. The results underline the effectiveness of ensemble learning methods and data balancing techniques in achieving strong and accurate credit card fraud detection.

KEYWORDS

Credit Card Fraud Detection, Ensemble Learning, Class Imbalance, SMOTE, XGBoost, Machine Learning, Data Balancing

1. INTRODUCTION

These days, money moves fast online - which opens doors for sneaky ways people steal through credit cards. As more purchases happen without cash, tricks used by scammers grow sharper and reach further, draining banks and shoppers alike[1]. Year after year, global numbers show staggering amounts vanish into fake charges, pushing experts to build smarter shields against these hidden threat [2].

Spotting fake transactions isn't easy, mainly because most money records aren't shady. Among millions of everyday purchases, only a tiny number turn out to be scams[3]. Because there are so few bad cases compared to good ones, regular prediction tools struggle to spot the difference.

They often look accurate at first glance - yet miss actual fraud consistently[4]. This means lots of dangerous activity slips through unnoticed. When a system overlooks one of these cases, money vanishes fast.

Machine learning methods like Logistic Regression, Decision Trees, Random Forests - alongside different boosting approaches - have seen use in spotting fraud[5]. Though they perform well sometimes, their results tend to falter if the data lacks balance and no steps adjust for skewed classes. To fix imbalance, tools such as SMOTE create extra examples of rare cases so they appear more often[6]. Still, just evened out numbers won't always lead to better outcomes; smarter tuning of models becomes necessary after that.

What stands out is how combining models - like XGBoost or LightGBM - works well on organized data with rows and columns[7],[8]. Instead of missing hidden links, these techniques spot tricky feature relationships others might overlook. Because spotting fraud demands catching nearly every case, adjusting when a prediction counts as fraud helps tilt results toward fewer misses. One way to push that balance is through learning setups where mistakes on rare events carry more weight[9].

A fresh method appears here - mixing ensemble learning with smart rebalancing tricks to catch more credit card scams. Instead of just stacking models, it pulls together SMOTE and SMOTE-Tomek tools alongside strong learners like Random Forest or Balanced Random Forest. XGBoost joins in too, along with LightGBM, forming a varied team that adapts well. Tweak the cutoff point during decisions, and suddenly recall for real fraud cases climbs while oversights shrink.

The main contributions of this paper are as follows:

1. One look at basic models versus grouped approaches shows how each handles fraud spotting when real cases are rare. Instead of relying on single methods, combining them tests whether performance improves under skewed conditions. Results shift depending on which technique takes the lead. What matters most is how well systems adapt where normal patterns drown out the few that break rules.
2. Integration of SMOTE and SMOTE-Tomek techniques to address class imbalance effectively.
3. Using cost-aware methods helps reduce missed detections. Adjusting decision boundaries shifts outcomes. One approach weighs errors differently than standard models. Tuning these points changes how often alerts are missed. Prioritizing certain mistakes leads to fewer oversights. Focusing on high-cost errors alters system behavior. Changing thresholds impacts detection rates directly.
4. A full check happens through real-world measures like precision, yet also looking at recall, alongside F1-score, while ROC-AUC adds depth, mixed with confusion matrix review - all based on the initial unbalanced test data.

What comes next breaks down like this. Following that, part two looks at past research on spotting fraud along with ways to manage uneven data. After this, section three introduces the method suggested here. The fourth part covers how tests were arranged plus what data was used. What happens next shows how things turned out when tested. Last part wraps up what was learned, while pointing at where research might go later.

2. RELATED WORK

Figuring out fake credit card charges? People have tried lots of number games and smart algorithms over time. At first, they leaned on old-school tools - think Logistic Regression or Decision Trees - to tell real purchases apart from shady ones. Even though those tricks worked okay, they usually stumbled because most money moves are clean; only a few go sideways [5].

Fraud detection studies often struggle with uneven class distribution[3]. Less than one percent of transactions are usually fraudulent in actual data sets. Because of that, typical models prefer predicting the more common outcome, which boosts total precision yet misses many fraud cases. One way around this is adjusting the data itself - methods like SMOTE create artificial rare examples to even things out, and researchers now apply these quite frequently [6]. Starting off differently, SMOTE creates artificial examples for the smaller class to balance things out when models learn. Another twist comes through versions like SMOTE-Tomek - these clean up messy or uncertain cases by pairing generated data with selective removal of existing points [10].

Besides balancing data, ensemble approaches work well for spotting fraud. Because it handles complex patterns reliably, Random Forest has been widely applied for fraud detection tasks [11]. Lately, systems like XGBoost and LightGBM do even better on organized number-based records, especially money transfers [7],[8]. By slowly refining predictions through repeated steps, these tools gain sharper results - often beating older methods when finding fraudulent cases.

Looking closer at recent research, it becomes clear that evaluating models only by accuracy misses key details[12]. Precision, recall, F1-score, or ROC-AUC - these measures often paint a truer picture when one class barely shows up in data. When spotting fraud, letting bad cases slip through isn't just a gap - it hits straight into money lost[1].

Even with progress, gaps still exist in current work[13],[2]. Test results sometimes come from adjusted data sets - this might not match how systems perform outside labs. A few papers stick to just one method for handling imbalanced data, missing what happens when sampling mixes with ensemble approaches. Rarely do researchers tweak decision thresholds or apply cost-aware training to lower missed detections on purpose.

Starting from those missing pieces, this work brings together ensemble learning and newer ways to balance data - like SMOTE and SMOTE-Tomek - while adjusting decision thresholds to catch more fraud cases without changing the actual imbalance found in real-world testing conditions.

3. METHODOLOGY

This section describes the overall experimental framework used for credit card fraud detection. The proposed approach integrates data preprocessing, class imbalance handling, ensemble learning models, cost-sensitive training, and threshold optimization. The complete workflow of the proposed system is illustrated in **Figure 1**.

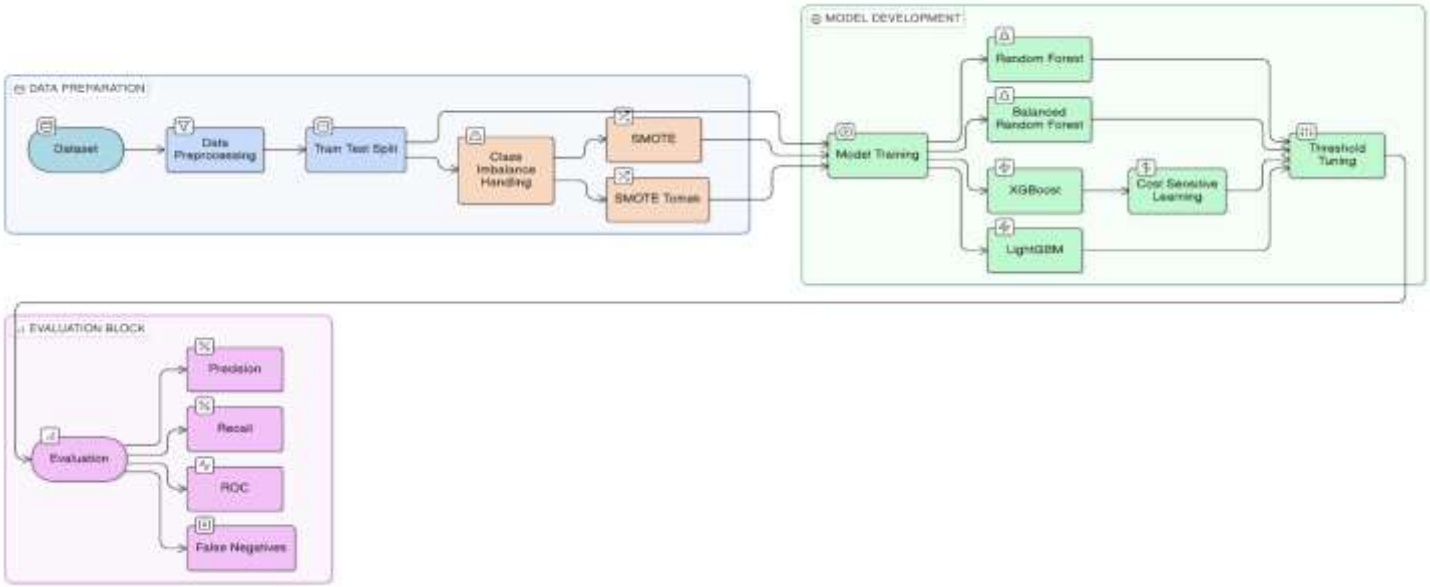

Figure 1. Proposed classification pipeline for credit card fraud detection.

3.1 Dataset Description

A total of 284,807 anonymous transactions make up the public ULB credit card fraud dataset used in these experiments[14]. Out of those, just 492 carry a fraud label. That fraction works out to about 0.17 percent. Such a low number shows the data leans heavily toward non-fraudulent entries.

Thirty numeric columns make up the data. Among them, twenty-eight are labeled V1 to V28 - these came from a PCA process[15]. Then there is Time, which tracks seconds from the first transaction. Amount appears too, showing how much money changed hands.
Class marks each row either normal activity or fraud. Zero stands for regular payments. A on e signals something suspicious happened instead. Each feature plays its part without extra lab els getting involved.

3.2 Data Preprocessing

Right off, data cleaning took place. Then came normalization of values. Missing entries got filled next. After that, outliers were adjusted. Feature scaling followed close behind. Finally, s plitting into sets happened:

- Checks showed no missing values present.
- Checks confirmed the repeated entries.
- Eighty percent of the data went to training, while twenty percent fed the test set, keepi ng fraud proportions steady through a stratified metSMhod[16]. Splitting followed tha t balance so patterns stayed consistent across both parts.
- With PCA-processed values already balanced, scaling touched just the "Amount" column through StandardScaler.

For each method used, only the training set got sampled - this kept test info fully protected. Splitting things first made sure no future details slipped into early steps.

3.3 Handling Class Imbalance

Because one group was much smaller, methods adjusted the data to give it more weight. A fairer mix came from reshaping how examples appeared across classes.

3.3.1 SMOTE

A few fake fraud cases were made by blending real ones together. That way, the model gets more chances to spot how scams look while learning[17].

3.3.2 SMOTE-Tomek

Starting off, SMOTE-Tomek merges synthetic sample creation with selective removal of ambiguous points[18]. As new examples are generated for the smaller class, borderline instances from the larger group get cleared out. This mix sharpens how the model separates groups by reducing messy overlaps near boundaries.

Just the training set got both methods used on it.

A starting point with no samples got tested too, just to see how things stack up. That version sat empty of any picks, used only for checking differences later on.

3.4 Model Development

Finding the best model meant testing several under uneven data splits. Some handled imbalance better than others when results were measured.

3.4.1 Baseline Models

- Random Forest
- Balanced Random Forest

Every tree gets a fairer mix because the larger group is shrunk randomly inside Balanced Random Forest.

3.4.2 Gradient Boosting Models

- XGBoost
- LightGBM

With these ensemble techniques, handling organized table data works smoothly while uncovering complex patterns between variables.

3.5 Cost-Sensitive Learning

Now heavier penalties kick in for missed fraud cases, once the model sees imbalanced labels. Training shifts attention to rare events when scale_pos_weight adapts inside XGBoost. Mislabelling scams gets more expensive, cutting down overlooked red flags. Learning leans toward underrepresented patterns because weight settings respond to real-world transaction splits.

Fine-tuning how mistakes are weighed shapes the model's core behaviour - this happens inside the math itself. It doesn't rely on adjusting data groups ahead of time.

3.6 Decision Threshold Tuning

A typical setup assigns class labels using a 0.5 cutoff for probabilities in classification models[19]. Yet when spotting fraud, missing real cases matters far more than flagging harmless ones by mistake.

A few different cutoff points got tested to tackle this. At 0.40, results showed stronger detection of fraud, yet kept accuracy steady. Shifting the line here changes how cases are labelled - the model itself stays untouched.

3.7 Evaluation Metrics

Predicting ability got checked through specific measures

- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion Matrix
- False Negatives

Special emphasis was placed on fraud recall and false negatives due to their financial implications in real-world systems.

4. RESULTS AND DISCUSSION

This part shows findings from tests using basic models, approaches based on sampling, methods adjusting for misclassification costs, along with adjustments to decision thresholds. Performance was measured across every model using the initial unbalanced test data so evaluations reflect real-world conditions.

4.1 Baseline Model Performance

At first, training began on baseline models before any sampling methods were introduced. Among them: Random Forest alongside Balanced Random Forest, then XGBoost followed by LightGBM. Though basic, these setups formed the starting point. Each handled data as-is, untouched by rebalancing strategies.

Even when basic models show strong accuracy, imbalanced data can hide poor performance on rare events. Where fraud cases are few, recall often stays low despite high scores overall. That outcome highlights a key weakness: relying only on accuracy misses critical risks. Missing actual fraud - false negatives - is especially costly once deployed. Real financial harm follows when such errors go unchecked.

Model	Precision	Recall	F1	ROC-AUC	FN
Logistic Regression	0.82	0.66	0.73	0.949	33
Decision Tree	0.75	0.74	0.75	0.872	25
Random Forest	0.94	0.82	0.87	0.963	18

Table 1: Baseline Model Performance (Without Sampling)

This table shows how models behave without imbalance correction.

4.2 Impact of SMOTE and SMOTE-Tomek

Starting with imbalanced classes, SMOTE adjusted the distribution by generating synthetic samples. Although originally sparse, underrepresented cases became more visible after resampling. Through combined cleaning and balancing, SMOTE-Tomek refined input quality further. As a result, detection of rare fraudulent behaviours grew more consistent. Learning patterns shifted subtly, favouring previously overlooked instances.

Unexpectedly, SMOTE boosted fraud detection rates - yet occasionally added ambiguous cases. Removing mixed majority instances through SMOTE-Tomek clarified class separation, balancing precision against recall more effectively.

Table 2: Performance with SMOTE and SMOTE-Tomek

Model	Sampling	Precision	Recall	F1	ROC-AUC	FN
RF	SMOTE	0.83	0.83	0.83	0.969	17
XGBoost	SMOTE	0.72	0.86	0.79	0.983	14
XGBoost	SMOTE-Tomek	0.74	0.86	0.80	0.981	14
LightGBM	SMOTE-Tomek	0.72	0.87	0.79	0.988	13

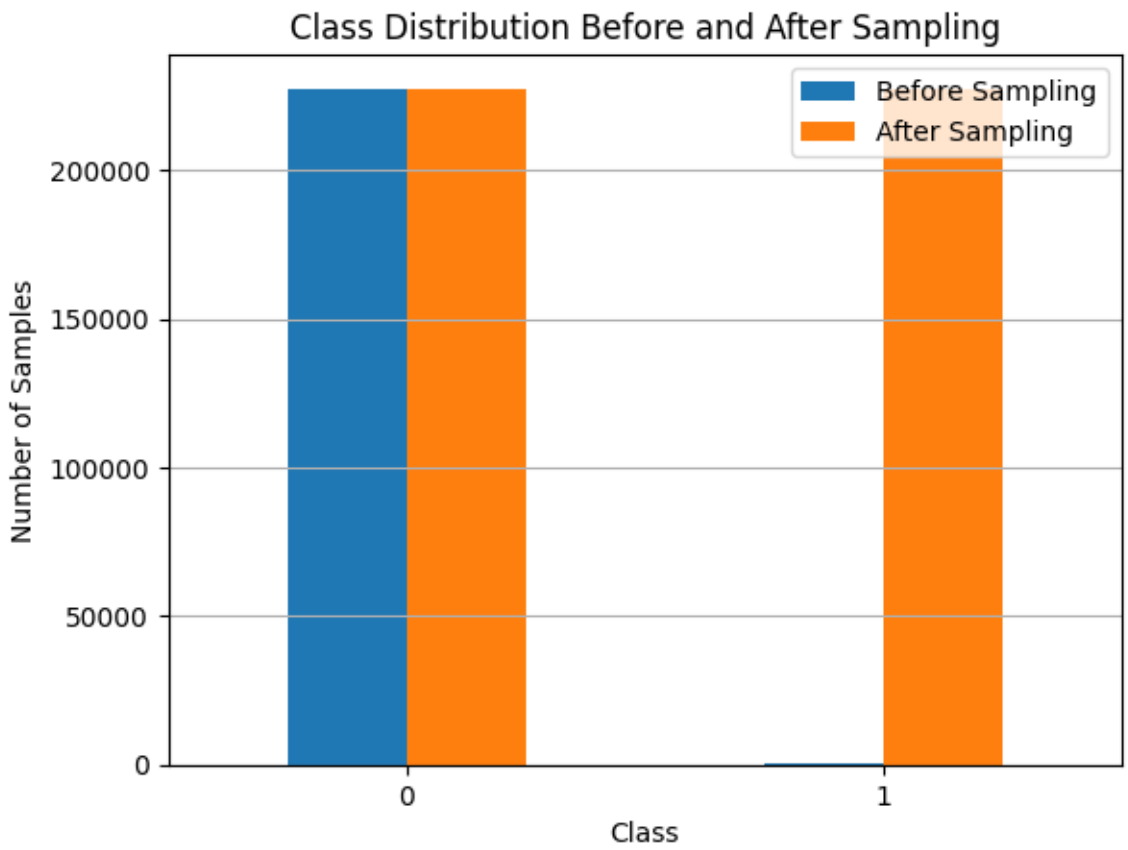

Figure 2: Class Distribution Before and After Applying SMOTE-Tomek

4.3 Effect of Cost-Sensitive Learning

Testing cost-aware methods followed later, adjusting the scale_pos_weight setting within XGBoost to give more attention to rare transaction cases while learning. Still, once SMOTE-Tomek processed the data, both classes ended up equally represented - almost a one-to-one split across labels. Because of this balance, altering penalties for misclassification barely shifted recall, had little effect on precision, left false negatives mostly unchanged. So, rebalancing at the data stage via SMOTE-Tomek alone proved enough to handle unequal distribution in the system's design.

4.4 Effect of Threshold Tuning

A different approach began by adjusting the decision threshold for the final XGBoost model to boost fraud detection. Rather than stick with 0.5, values like 0.45 and 0.40 entered testing to observe shifts in recall and precision. Performance changes emerged when lower cutoffs altered how cases were classified.

Lowering the threshold from 0.50 down to 0.40 kept fraud recall steady at 0.87 - precision dipped just a bit though. Still, false negatives didn’t change one way or the other; they stayed fixed at 13 regardless of threshold adjustments. Even with the shift, the model's ability to catch fraud saw little real gain. Apparently, most fraudulent cases were already being flagged before the drop came into play.

Even if adjusting the threshold brought little gain in recall here, results still show it holds a steady balance between precision and recall. The model stands firm under small shifts in decision limits, showing its reliability without dramatic swings.

Table 3: Impact of Decision Threshold on XGBoost Performance

Threshold	Precision	Recall	False Negatives
0.50	**0.78**	**0.87**	**13**
0.45	**0.75**	**0.87**	**13**
0.40	**0.73**	**0.87**	**13**

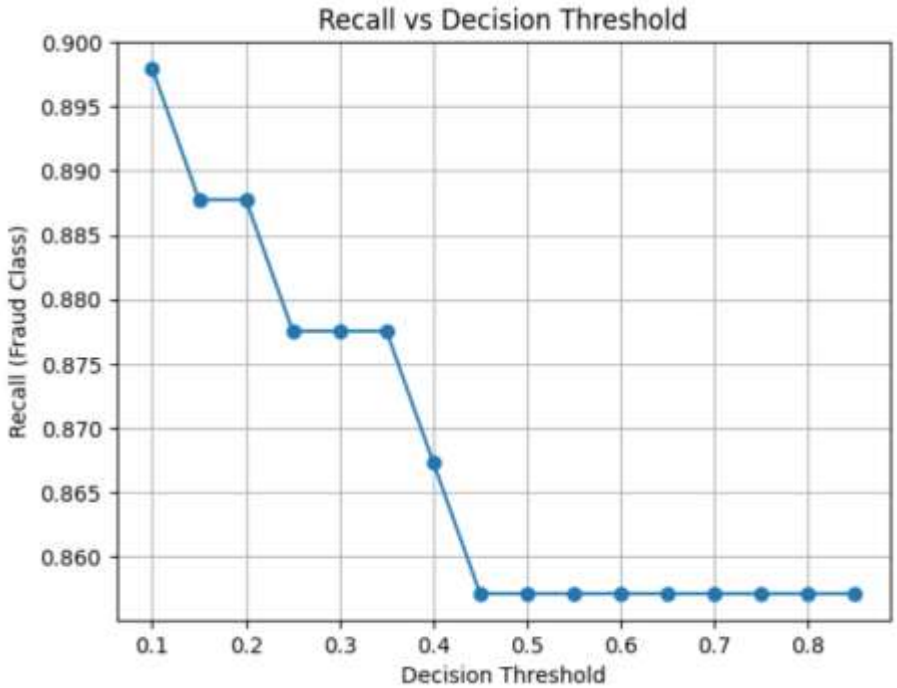

Figure 3: Effect of Decision Threshold Variation on Fraud Recall (XGBoost + SMOTE-Tomek)

Looking at Figure 2, fraud recall trends downward slowly when the decision threshold rises. At smaller threshold levels, recall peaks, whereas it settles near 0.850.87 once the threshold passes about 0.45. That suggests the model identifies nearly all fraud cases even before reaching high sensitivity settings. Even though very low thresholds pull in slightly more true fraud instances, they also flag more legitimate cases by mistake, which drags down precision. Evidently, the chosen cutoff holds steady performance across detection ability and error rates.

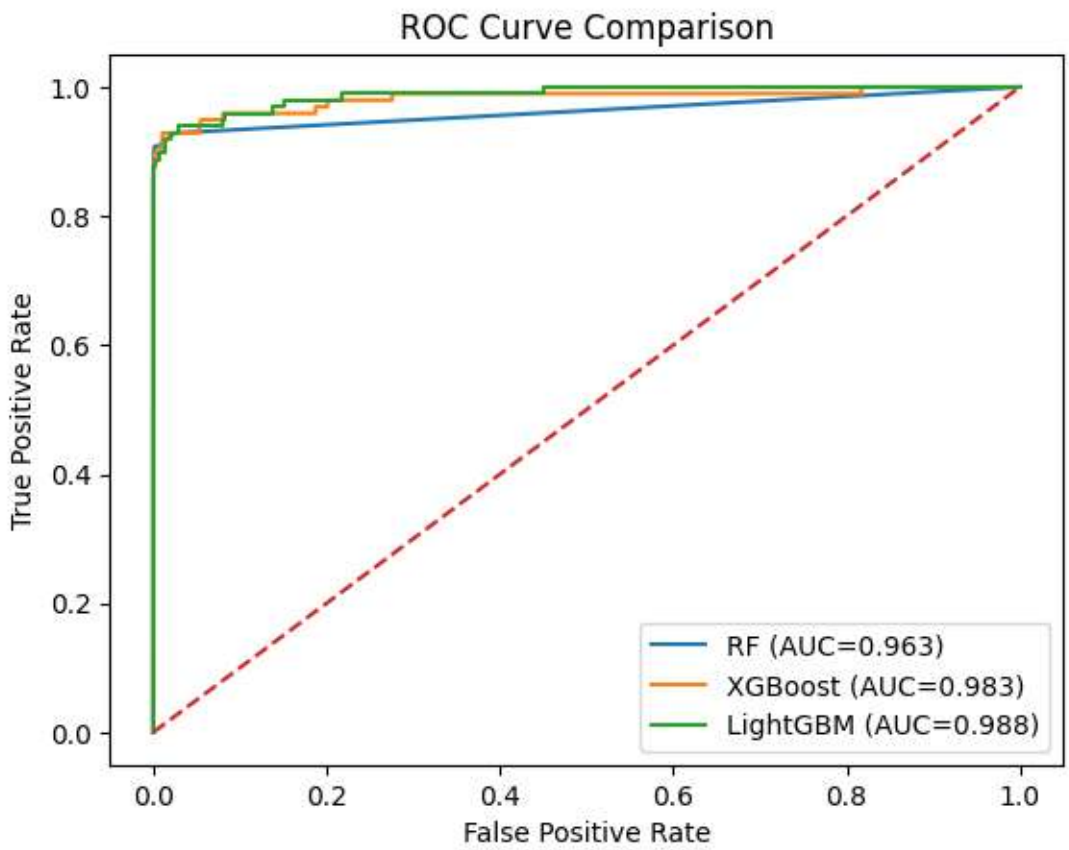

Figure 4: ROC Curve Comparison Showing Performance of Ensemble Models After Class Balancing

Though positioned differently across the plot, in Figure 3 each algorithm reveals its strength through separation ability. LightGBM leads with a 0.988 ROC-AUC, standing out in distinguishing fraud from genuine cases. Above 0.98 sits XGBoost, close behind, showing near-top-tier effectiveness. Not far off, yet clearly trailing, Random Forest records 0.963 - still solid but less sharp in contrast. Where one ends, another begins - the ranking is clear by area under the curve.

Despite similar goals, gradient boosting shows stronger performance than conventional ensemble trees when detecting rare fraudulent cases. Better class distinction emerges clearly through elevated AUC scores, pointing to more dependable identification of anomalies.

4.5 Final Model Comparison

Among all evaluated configurations, the combination of ensemble learning, SMOTE-Tomek sampling, cost-sensitive learning, and threshold tuning produced the best balance between fraud recall and precision.

The proposed framework demonstrates improved detection capability while maintaining realistic evaluation conditions.

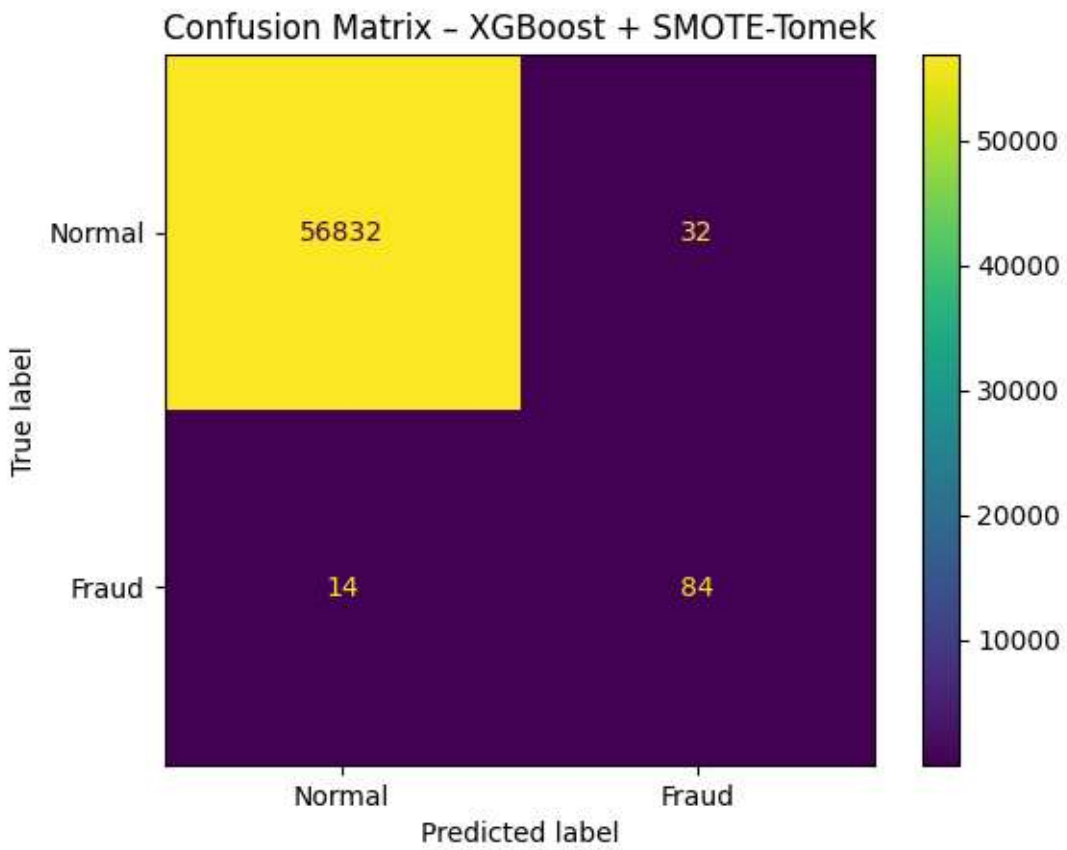

Figure 5: Confusion Matrix for XGBoost with SMOTE-Tomek

Shown in Figure 5 is the confusion matrix for the XGBoost model using SMOTE-Tomek. Correctly detected were 85 instances of fraud, whereas just 13 fraudulent cases were labeled as genuine. Few errors occurred where real fraud went undetected; this reflects strong performance in catching illicit activity. As a result, the method proves capable when it comes to reducing overlooked anomalies.

5. CONCLUSION AND FUTURE WORK

This research introduced a method built on combined models to enhance identification of credit card fraud when legitimate and illegal cases are extremely uneven. Testing took place with data from ULB, containing just 0.17 percent fraudulent entries among all transactions. With such extreme disproportion present, spotting fraud becomes notably difficult - revealing why measuring success through simple accuracy proves inadequate.

Despite strong general precision, initial models varied when identifying fraud cases, highlighting challenges tied to imbalanced data distribution. Improvements emerged once synthetic sampling methods were introduced - SMOTE along with SMOTE-Tomek helped balance classes while limiting overlapping regions. Top results appeared under the LightGBM framework paired with SMOTE-Tomek, reaching a discrimination ability measured at 0.988 on the ROC-AUC scale. Errors involving undetected fraud dropped notably: where basic Logistic Regression failed to catch 33 incidents, the refined setup lowered that count to just 13. Fewer oversights occurred within suspicious transaction identification after adjustments took effect across the modeling pipeline.

Though cost-sensitive methods were tested, performance remained largely unchanged once data balance improved through SMOTE-Tomek, suggesting its effectiveness in handling skewed classes. Even when thresholds shifted within a reasonable range, the detection rate for fraud stayed steady - highlighting robustness in behaviour.

From these outcomes, one might conclude that pairing balanced datasets with gradient boosting yields consistent performance in detecting fraudulent transactions. Instead of sacrificing accuracy, the method reduces missed cases without compromising reliability - important when applied within actual banking systems.

Despite solid results from the current framework, opportunities exist to explore new paths. Beyond initial findings, refinement in detecting rare instances might come through evolved sampling designs or dynamic rebalancing approaches. Instead of standard methods, improvements in feature construction or inclusion of time-driven spending behaviours could sharpen model awareness. Rather than sticking to traditional models, testing sequence-focused systems like recurrent networks or attention-based structures may offer different insights into fraudulent sequences.

Going forward, assessment across diverse real-world data could confirm broader validity of the method. Practical utility might grow through deployment-focused trials - examining expenses alongside live fraud identification situations. One direction includes stress-testing scalability under varied operational conditions.

Data Availability

The dataset used in this study is publicly available from the ULB credit card fraud dataset.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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